

LESSON 5-2 • SESSION 1 • PRACTICE • FORM B

Area of Triangles

Learning target – I can find the area of a triangle using one half the base times the height. • 6.G.A.1

REMEMBER

- Two matching triangles make a parallelogram, so a triangle is **half**: $A = \frac{1}{2} \times b \times h$.
- The **height** makes a square corner with the base.
- In a **right triangle**, the two legs at the square corner are the base and height.

WORKED EXAMPLE

A triangle has a base of 14 cm and a height of 9 cm. Find its area.

Step 1 $14 \times 9 = 126$ **Step 2** 126 is the parallelogram made by two triangles.**Step 3** $126 \div 2 = 63$ **Answer** 63 cm²**WORDS** right triangle (*triángulo rectángulo*) • height (*altura*) • congruent (*congruente*)

PRACTICE

show your work • label units

1. Find the area of each triangle with base
- b
- and height
- h
- .

a. $b = 16$ cm, $h = 5$ cm _____

b. $b = 9$ in, $h = 4$ in _____

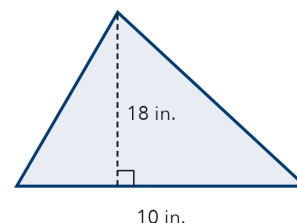
c. $b = 22$ ft, $h = 6$ ft _____

d. $b = 13$ m, $h = 8$ m _____

- 2.
- Pennant Flag.**
- Two copies of this flag fit together to make a parallelogram.

a. Area of the parallelogram _____ in²

b. Area of one flag _____ in²



- 3.
- Cut Sandwich.**
- A sandwich is cut on a diagonal into two equal triangles. Each has a base of 12 cm and a height of 9 cm.

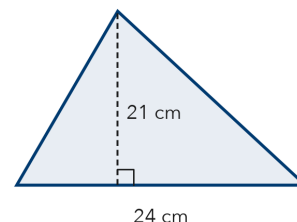
a. Area of one triangle _____ cm²

b. Whole sandwich _____ cm²

- 4.
- Right-Triangle Sail.**
- A sail is a right triangle. Its two legs are 8 ft and 15 ft. Find its area.

Hint The legs meet at a square corner.**Answer** _____ ft²

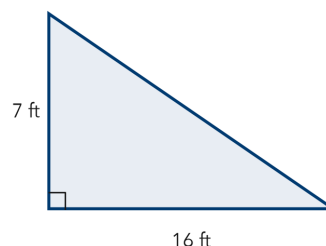
- 5.
- Warning Sign.**
- Find the area of the sign.

Hint Multiply, then take half.**Answer** _____ cm²

6. **Recess Wedge.** A wedge of the recess yard is a right triangle. Find its area.

Hint In a right triangle, the legs are the base and height.

Answer _____ ft²



7. **Two Banners.** Banner P is a triangle with base 16 m and height 9 m. Banner Q has base 12 m and height 14 m. Find each area. Which covers more space?

Show your work.

Areas, which _____

8. **Garden Plot.** A triangular garden has a base of 20 ft and a height of 18 ft. One bag of mulch covers 30 ft² and costs \$5.

- a. Find the area. _____ ft²
 b. How many bags are needed? _____
 c. Find the total cost. What happens to the cost if the base is doubled? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Marta's Work.** A triangle has a base of 20 m and a height of 7 m. Marta wrote: " $A = 20 \times 7 = 140$ m²." Explain her mistake. Then find the area.

TEST PRACTICE

choose the best answer

10. A triangle with base 16 m and height 5 m has area –
A 21 m² **B** 40 m² **C** 80 m² **D** 160 m²
11. The height of a triangle must meet the base at –
A a square corner (right angle) **B** the longest side
C the middle of the base **D** any angle

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

504 Q	\$30 T	112 Z	90 P	56 U	120 X	54 R	180 C	60 O	252 D
-----------------	------------------	-----------------	----------------	----------------	-----------------	----------------	-----------------	----------------	-----------------

#2b	#3a	#4	#5	#6	#8a	#8c
-----	-----	----	----	----	-----	-----

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____